Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| | | | | |
|---|---|---|---|---|
| **Proposer** | **Name** | Shubhra Bansal | **Email** | coria-consortium@purdue.edu |
| **Institute Name** | Purdue University | | | |
| **Address** | 585 Purdue Mall, West Lafayette, IN 47907-2088 | | | |

## 2. Technology Overview

| | | | | |
|---|---|---|---|---|
| **Title of Proposed Technology** | **Overcoming Commercialization-Stage Technical Challenges of High-Performance Perovskite Emitters through an AI-Based Optimization System** | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√ ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| **Industrial Technology Classification** [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Machinery and Materials* | | *Information & Communication* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Advanced Functional Materials* | *Display Emissive Materials* | *Artificial Intelligence* | *AI-based Materials Design & Optimization* |
| **Concept of Proposed Technology** | **Core Concept**<br>This technology proposes the development of an AI-assisted internal R&D tool designed to accelerate the company's perovskite emissive material development for next-generation display applications.<br>The platform integrates **material synthesis data, optical/electrical property data, reliability test results, and device-level performance data** into a unified research database. Rather than replacing experimental research, the AI system functions as a **decision-support engine** that shortens trial-and-error cycles, improves reproducibility, and maximizes the commercialization potential of core emissive materials.<br><br>**Key Functions**<br>Based on this data foundation, AI models are utilized to:<br>● Rapidly explore new perovskite compositions and structures | | | |

*(max 10 pages)*

| | |
|---|---|
| | ● Predict material performance and degradation behavior<br>● Optimize material–process–device parameters simultaneously |
| **Rationale for Joint R&D Project** | **Critical Technical Barriers**<br>Perovskite emissive materials offer ultra-high color purity, high absorption coefficient, and compatibility with thin color-conversion structures, making them strong candidates for next-generation displays.<br>However, the commercialization of next-generation display materials based on perovskite emissive technologies presents technical challenges that fundamentally differ from laboratory-scale research. As development progresses toward practical application and market deployment, the primary bottlenecks shift from material discovery to reproducibility, long-term reliability, and system-level optimization.<br>These challenges arise from the intrinsic characteristics of perovskite materials, including strong sensitivity to composition, processing conditions, and operating environments. In particular, performance degradation mechanisms often emerge only under extended thermal, optical, and environmental stress conditions relevant to commercial displays, making them difficult to identify and resolve through isolated or short-term experiments.<br>At the commercialization stage, material performance, process conditions, and device-level behavior become highly interdependent, forming a complex, multi-parameter problem space. Conventional trial-and-error–based R&D approaches face inherent limitations in efficiently addressing such complexity due to:<br>● The large number of interacting variables across material synthesis, processing, and device integration<br>● Limited availability of long-term reliability data at early development stages<br>● Difficulty in extracting generalizable design rules from fragmented experimental results<br>As a result, there is a growing need for data-driven and predictive methodologies that can systematically link material properties to process conditions and device performance.<br><br>**Why Global Collaboration?**<br>International joint research with overseas leading research institutes provides a critical pathway to overcoming these commercialization-stage technical barriers.<br>Overseas research institutes often possess advanced capabilities in:<br>● AI-based materials modeling and predictive analytics<br>● Large-scale research data structuring and management<br>● Statistical and machine-learning approaches for degradation and failure analysis<br>By integrating experimental data generated during material development with advanced modeling and analytical frameworks, joint research enables:<br>● Identification of key performance-limiting and degradation mechanisms<br>● Predictive optimization of material composition and processing parameters<br>● Reduction of uncertainty in scale-up and commercialization decisions |

*(max 10 pages)*

| | |
|---|---|
| | Such collaboration allows for a systematic transition from empirical development to predictive, data-driven R&D, which is essential for accelerating commercialization timelines. |
| **Key Words** | 1. Perovskite Emissive Materials<br>2. AI-driven Materials Discovery<br>3. Display Color Conversion<br>4. Research Data Platform<br>5. Commercialization Acceleration |

*(max 10 pages)*

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 4 ) | End of the Project | Level ( 6 ) |
|---|---|---|---|---|
| **R&D Objectives** | ● Establish a standardized research data acquisition and management system for perovskite emissive materials<br>● Develop AI models for predicting material properties, stability, and degradation trends<br>● Implement AI-assisted optimization of material composition and processing parameters<br>● Demonstrate accelerated development of high-performance, commercially viable perovskite emissive materials | | | |
| **R&D Target Specifications** | ● Development of AI models enabling data-driven new material exploration and process optimization<br>● Generation and experimental validation of two (2) AI-assisted process recipes for emissive material development<br>● Definition and implementation of five (5) standardized measurement and evaluation protocols, supported by official test or verification documentation | | | |
| **R&D Plan** | **Phase 1: Establishment of Research Data Infrastructure**<br>● Definition of key material, process, and device parameters<br>● Standardization of experimental protocols and data formats<br>● Construction of integrated research database<br><br>**Phase 2: AI Model Development**<br>● Development of machine-learning models linking composition, process, and performance<br>● Implementation of degradation and reliability prediction models<br><br>**Phase 3: AI-Assisted Material Optimization**<br>● AI-guided screening of candidate materials<br>● Iterative experiment–model feedback loop<br><br>**Phase 4: Prototype-Level Validation**<br>● Fabrication and evaluation of optimized emissive material prototypes<br>● Verification of performance and scalability for display applications | | | |

*(max 10 pages)*

## 4. R&R (Roles and Responsibilities)

| | |
|---|---|
| **Expected R&R of Research Institute** | ● Development of AI algorithms and modeling frameworks<br>● Construction and operation of research data infrastructure<br>● Advanced data analysis and performance prediction<br>● Joint interpretation of results and model refinement |
| **Expected R&R of Korean SME Partner** | ● Provision of perovskite emissive material synthesis and processing expertise<br>● Generation of high-quality experimental and reliability data<br>● Prototype fabrication and performance validation<br>● Commercialization-oriented specification definition and feedback |

*(max 10 pages)*

## 5. Expected Outcomes/Impact

| | |
|---|---|
| **Expected Outcomes and Impact** | This joint R&D project is expected to:<br>● Establish a scalable AI-based material R&D paradigm for display applications<br>● Significantly reduce development time and cost for next-generation emissive materials<br>● Enhance the commercial readiness and competitiveness of perovskite-based display materials<br>● Provide a transferable framework applicable to other advanced nanomaterials and device platforms<br>Ultimately, the project will contribute to strengthening Korea's **core material technology leadership** in the global display industry |

## 6. References

(1) H. Cho+, S.-H. Jeong+, M.-H. Park+, T.-W. Lee* et al., *Science*, 350, 6265, 1222-1225 (2015).

(2) Y.-H. Kim+, S. Kim+, A. Kakekhani+, A.M. Rappe*, T.-W. Lee* et al., *Nature photonics*, 15, 2, 148-155 (2021).

(3) Y. -H. Kim+, J. Park+, S. Kim+, T. -W. Lee* et al., *Nature Nanotechnology*, 17, 6, 590-597 (2022)

(4) J. S. Kim+, J. -M. Heo+, T. -W. Lee* et al., *Nature*, 611, 688–694 (2022).

(5) H.-D. Lee+, S.-J. Woo+, S. Kim+, T.-W. Lee* et al., *Nature Nanotechnology*, 19, 5, 624–631 (2024)

(6) Q. Zeng+, T.-W. Lee* et al., *Science*, 391, 6782 (2026).